

BIPV Digital Twin: Full Technical & AI Documentation

1. Executive Summary

This project develops an AI-driven "Digital Twin" for Building-Integrated Photovoltaics (BIPV). By combining machine learning with real-world solar physics, the system provides high-fidelity energy yield predictions that allow architects and engineers to validate solar designs before construction. The core value lies in bridging architectural concepts with data-backed engineering evidence through an automated simulation pipeline.

2. Technical Architecture & Data Pipeline

The system is built on a modular "Inference Pipeline" that manages the complete lifecycle of a simulation request:

- **Coordinate Retrieval:** The system uses **geopy** to translate user-inputted city names into precise Latitude and Longitude coordinates.
- **TMY Data Fetching:** The pipeline queries localized **Typical Meteorological Year (TMY)** datasets to retrieve hourly atmospheric conditions.
- **Solar Geometry Logic:** Powered by **pvlb**, this layer calculates the **Angle of Incidence (AOI)** by processing surface tilt, azimuth, and the sun's position relative to the building at each hour of the year.
- **Backend Infrastructure:** A **FastAPI** server coordinates these micro-services, executing the AI inference and returning structured results.

3. XGBoost AI Model: Feature Breakdown

The primary predictive engine is an **XGBoost** model (loaded via **joblib**) trained on \$8,760\$ hourly rows of data. It utilizes a multidimensional feature set to forecast power output (\$W/m^2\$):

- **Atmospheric Variables:**
 - **GHI (Global Horizontal Irradiance):** Total solar radiation on a horizontal surface.
 - **DNI (Direct Normal Irradiance):** Solar radiation coming directly from the sun.

- **DHI (Diffuse Horizontal Irradiance):** Solar radiation scattered by the atmosphere.
 - **Air Temperature ($^{\circ}\text{C}$):** Used to model voltage drops and thermal efficiency losses.
 - **Wind Speed (m/s):** Used to determine the convective cooling effect on the BIPV modules.
 - **Geometric & Temporal Variables:**
 - **AOI (Angle of Incidence):** The specific angle at which sunlight strikes the panel surface.
 - **Surface Tilt & Azimuth:** The fixed physical orientation of the BIPV module.
 - **Hour/Month:** Encodes seasonal and diurnal patterns.
-

4. Modeling Assumptions & Guardrails

To ensure conservative and reliable energy estimates for professional use, the following parameters are applied:

- **Performance Ratio (Losses):** A **14% system loss factor** is included to account for real-world environmental factors like soiling, wiring, and inverter inefficiency.
 - **Fixed Efficiency:** Predictions utilize fixed conversion rates based on the selected material (e.g., **22%** for Monocrystalline, **12%** for Thin Film).
 - **Orientation Persistence:** The model assumes static building integration (no tracking) to reflect standard architectural practices.
 - **Typical Climatology:** Uses TMY data to represent "typical" long-term expectations rather than short-term weather anomalies.
-

5. API Validation (FastAPI Swagger UI)

The project's functionality is validated through a **FastAPI Swagger (OpenAPI) UI**. This interface proves the "brain" of the Digital Twin can process complex requests and return real-time analytics.

Sample Request Parameters:

- **City:** (e.g., Berlin)
- **Panel Area:** Total square meters
- **Material ID:** (e.g., Standard Panel)

Sample JSON Output: The API returns a structured dataset including **Yearly Total Yield (kWh)**, **Monthly Generation Curves**, and **Weather Triggers** (percentage of sunny vs. cloudy days).

6. Future Roadmap

- **Interactive 3D Builder:** Developing a **React-Three-Fiber** interface for modular block assembly.
- **Live Sensor Integration:** Transitioning from TMY climatology to real-time IoT weather feeds.
- **Material Library Expansion:** Adding semi-transparent and colored BIPV glass with varying transparency-to-efficiency ratios.